

Yang Huang

Machine Learning / Operations Research Algorithm Intern

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Education

Institute of Automation, Chinese Academy of Sciences

Ph.D. Candidate, direct Ph.D. track

2022–Present

Beijing

Research focus: generative models, reinforcement learning, and discrete optimization.

University of Science and Technology Beijing

B.S. in Mathematics and Applied Mathematics

2018–2022

Beijing

Mathematics background; recipient of Outstanding Student Scholarships in multiple academic years.

Publications

ICML 2026 First Author. Optimal Transport–Guided Stochastic Control for Graph Combinatorial Optimization. *International Conference on Machine Learning (ICML 2026)*, Regular

TMLR First Author. Learning to Destroy and Repair Local Optima in Multilinear Graph QUBO Optimization. *Transactions on Machine Learning Research (TMLR)*, Under Review

Research Experience

OT-Guided Diffusion and Stochastic Control for Graph Combinatorial Optimization Jan. 2025–Sep. 2025
First Author / Lead Researcher Accepted by ICML 2026; [Code](#)

- Unified MIS, MaxCut and MaxClique as QUBO problems under a multilinear relaxation on $[0, 1]^n$; proved no optimality gap and the existence of binary optimal solutions.
- Improved diffusion-style neural combinatorial optimization by replacing hand-designed or score-only sampling dynamics with an OT-guided controlled diffusion toward $\pi^*(x) \propto \exp(-f(x; G))$.
- Derived the optimal drift from the entropic OT / Schrödinger bridge formulation and made it tractable via Laplace approximation, yielding a proximal energy-minimization target amortized by a GNN controller.
- Trained the controller from rollout-buffer states with Laplace-energy supervision; performed multi-particle Euler–Maruyama sampling and greedy decoding for feasible discrete solutions.
- Achieved average MIS sizes of 20.12/43.41 on RB-small/ER-large, improving over the best learning baseline by 0.50/0.07; achieved MaxCut values of 731.01/2965.94 on BA-small/BA-large, improving over DiffUCO by 3.69/18.41.

Learned Destroy-and-Repair Test-Time Search for Multilinear Graph QUBO Oct. 2025–May 2026
First Author / Core Algorithm Designer Submitted to TMLR; [Code](#)

- Studied nonconvex local optima in multilinear QUBO relaxations, where direct gradient optimization is slow and amortized neural solvers can repeatedly converge to suboptimal binary solutions.
- Designed a recurrent repair network as a learned energy-based optimizer, minimizing the original relaxed QUBO energy from random or damaged continuous states before discrete decoding.
- Designed a destroy policy for test-time refinement: it observes graph structure, current state and search step, then converts node-level probabilities into a learned destruction rate for randomized masking.
- Trained repair and destroy alternately using random damage, policy-gradient rewards from decoded-score improvement, and repair fine-tuning over full destroy-repair trajectories.
- With the same 512 repair-call budget, improved over repair-only by 2.16/5.91 MIS vertices and 2.06/11.32 MaxCut value on small/large graphs.

Relational Attention Learned Optimizer for Asymmetric TSP May 2026–Present
Ongoing Research / Initial Algorithm Design Research in Progress

- Study Asymmetric TSP with a general directed cost matrix, where neither symmetry nor triangle inequality can be assumed, by optimizing in the city-position assignment space instead of directly generating tours.
- Represent a tour by assignment variables $x[i, t]$, where city i is placed at position t ; express the ATSP cost as $\sum_t \sum_{i,j} C[i, j]x[i, t]x[j, t + 1]$, connecting the problem to assignment-QUBO, multilinear relaxation, and Sinkhorn projection.
- Designed a relational attention optimizer that treats each (i, t) assignment variable as a token and explicitly models forward transition, backward transition, same-city, and same-position relations.
- Formulated the model as a learned optimizer: given the cost matrix and current relaxed assignment, it outputs an improved soft assignment trained with relaxed ATSP objective, row/column constraint penalties, binary regularization, and Sinkhorn constraints.
- Exploring a two-stage untied unrolled optimizer, where stage0 constructs a coarse assignment from random logits and stage1 refines it with separate parameters.

Skills

Programming	Python, PyTorch, DGL, NumPy, NetworkX; GPU training, batched evaluation and experiment logging
ML / RL	Graph neural networks, policy gradient, stochastic differential equation sampling, multi-particle inference
Optimization	QUBO, multilinear relaxation, optimal transport / stochastic control, local search, Sinkhorn projection, MIS / MaxCut / MaxClique / ATSP
Tools	Linux, tmux, Git, LaTeX; ablation design and paper reproduction